Senior Software Engineer, Decision Science

KoBold Metals · Remote

Job ID (this archive)	ENERGY-103
Employer	KoBold Metals
Location(s)	Remote
Career stage (archive tag)	Experienced (Senior)
Location group	Other U.S.
Majors (archive tags)	Data Science, Computational & Geospatial Science
Energy area	Critical minerals / data science
Pay listed	$170,000 – $215,000 per year
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/koboldmetals/jobs/4624637005
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-074

Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

Senior Software Engineer, Decision Science

About the Company

In the coming decades, demand for critical minerals such as copper, lithium, and cobalt will continue to grow. These minerals are needed to build electric infrastructure, vehicles, data centers, and electronics. However, the mining industry is making 10x fewer discoveries per dollar spent on exploration compared to 30 years ago because the easy-to-find, near surface deposits have largely been discovered.

KoBold was founded to be the first AI native mineral exploration, mine development, and technology development company. Our goal is to turn mineral exploration into a repeatable science to discover the minerals the world needs—by integrating artificial intelligence, human intelligence, and new technology. Our multidisciplinary teams of data scientists, engineers, and geologists work together to build data systems, design new sensors, and establish industry-leading workflows to develop and test subsurface predictions on our exploration projects.

Our first discovery, Mingomba, is a high-grade copper deposit in Zambia, found after spending materially less than industry average. We are developing Mingomba with the same technical rigor that we apply to exploration. The Mingomba team brings together mine and processing engineers, geologists, data scientists, software engineers, and community, health & safety, construction, and operations specialists. By combining industry knowledge and cutting edge technology, we are developing Mingomba into a world class project that has safer and more predictable outcomes for ourselves, partners, and communities. It is expected to create thousands of good jobs for generations.

KoBold runs exploration programs in eight countries, all with the potential for another discovery.

KoBold is privately held; investors include institutional asset managers T. Rowe Price and Canada Pension Plan Investments; technology venture capitalists Andreessen Horowitz, Breakthrough Energy Ventures, BOND Capital, Durable Capital, StepStone, and Standard Investments; and strategic natural resources companies BHP and Mitsubishi.

About the Role

At KoBold we believe that a modern scientific computing stack will enable systematic mineral exploration and materially improve our rate of mineral discovery. This role is a key ingredient to this strategy. As a member of our scientific computing team, you will apply decision science techniques in order to build scalable systems to help make high-speed, high-quality decisions for our mineral exploration projects. Collaborating with our exceptional team of data scientists, software engineers,

and geologists, you will tackle complex scientific problems head-on and collectively pave the way for discoveries of vital energy transition metals like lithium, copper, nickel, and cobalt. Together we can shape the future of mineral exploration and contribute to building a sustainable world.

Responsibilities

- Architect, implement, and maintain decision science libraries that will be used in KoBold's mineral exploration analyses.
- Build tooling to increase the velocity of our decision making, including enabling rapid prototyping in Jupyter notebooks; build experimentation, evaluation, and simulation frameworks; turning successful R&D into robust, scalable pipelines; and organizing ML models and their outputs for repeatability and discoverability.
- Apply–and coach team members to use–engineering best practices such as writing robust, testable and composable code
- Collaborate with data scientists, geoscientists and engineers to invent the modern decision science technology for mineral exploration
- Occasional travel to exploration sites around the world to observe the impact of scientific computing on KoBold's exploration products and design new technologies to further discovery. Travel is approximately twice per year depending on project needs.

Qualifications

Our ideal candidate will have:

- At least 5 years of experience in the field of decision science with a strong software engineering focus, though most great candidates will have closer to 10.
- Track record of building production quality data processing solutions or tooling that have delivered business value
- Proficiency with foundational concepts of ML, including statistical, traditional and deep-learning approaches
- Proficiency in Python, ideally including array-based packages such as xarray and numpy
- Deep experience with measured scientific data
- Experience in visualizing scientific data for domain experts
- Experience in MLops and in the making of robust ML systems
- Drive to increase the velocity and effectiveness of our data scientists in both experimental and production workflows
- Capacity to dive deep on novel challenging problems in applying decision science to mineral exploration, including understanding a complex domain of geology and mineral exploration practices as well as working with limited, disparate and noisy data sources
- Collaborative attitude to work with stakeholders with different backgrounds (data scientists, geoscientists, software engineers, operations)

Work practices and motivation:

- Ability to take ownership and responsibility of large projects.
- Intellectual curiosity and eagerness to learn about all aspects of mineral exploration, particularly in the geology domain. Open to working directly with geologists in the field. Enjoys constantly learning such that you are driving insights and innovations.
- Ability to explain technical problems to and collaborate on solutions with domain experts who aren't software developers. A strong communicator who enjoys working with colleagues across the company.
- Excitement about joining a fast-growing early-stage company, comfort with a dynamic work environment, and eagerness to take on a range of responsibilities.
- Keen not just to build cool technology, but to figure out what technical product to build to best achieve the business objectives of the company.
- Ability to independently prioritize multiple tasks effectively.

What to Expect

Joining KoBold means getting the opportunity for hands-on exposure to our exploration projects around the world. All employees are expected to travel to project sites, with a minimum of one week per year. Field-facing and technical roles spend significantly more time in the field.

KoBold Metals is an equal opportunity workplace and an affirmative action employer. We are committed to equal employment opportunity for people of any race, color, ancestry, religion, sex, gender identity, sexual orientation, marital status, national origin, age, citizenship, disability, or veteran status.

This position is Full-time

The US base salary range for this full-time exempt position is $170,000 - $215,000

Location: Remote, Candidates can be located anywhere in the United States or Canada. All candidates must be legally authorized to work in the United States or Canada.

Compensation ranges are listed in USD. If you are based outside the US, the range may not translate directly: US pay reflects a different market with different competitive dynamics, so we benchmark roles against the local market rather than converting the listed figure.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: In the coming decades, demand for critical minerals such as copper, lithium, and cobalt will continue to grow. These minerals are needed to build electric infrastructure, vehicles, data centers, and electronics. However, the mining industry is making 10x fewer discoveries per dollar spent on exploration compared to 30 years ago because the easy-to-find, near surface deposits have largely been discovered. | At least 5 years of experience in the field of decision science with a strong software engineering focus, though most great candidates will have closer to 10.

Key skills listed in the archive: Python; R

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 4624637005

Career stage: Senior

Major fit: Direct skills pathway: data science, machine learning, analytics or data engineering; exact degree/experience requirements retained. Energy-sector connection: Computational exploration and decision science for minerals used in electrification.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 4624637005.

Sources and retrieval details

Employer posting

https://job-boards.greenhouse.io/koboldmetals/jobs/4624637005

Retrieved source

https://boards-api.greenhouse.io/v1/boards/koboldmetals/jobs?content=true

Retrieved: 2026-10-26T16:33:09.479714+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.